The AI Lemons Problem

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AI can improve prediction while undermining market viability

Higher AI penetration widens spreads and drives out uninformed traders.

- The informed share of order flow rises with AI adoption
- Uninformed participation is endogenous to the spread

Takeaway

The market can collapse even as conditional prediction improves.

Prediction markets need counterparties, not information alone

The paper makes participation, rather than price accuracy alone, the central margin.

- Liquidity traders and recreational users sustain two-sided order flow
- Their willingness to trade falls when spreads become costly

Takeaway

Market composition determines whether information aggregation remains possible.

AI ADOPTION CREATES A SELF-REINFORCING ADVERSE-SELECTION SPIRAL

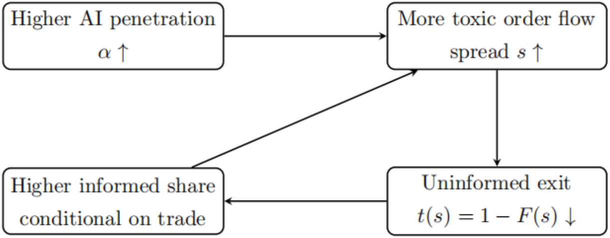

Figure: AI adoption raises informational asymmetry, spreads, and uninformed exit.

Participation feedback turns asymmetry into possible collapse.

The model combines informed trading with endogenous participation

- Binary prediction contract with state V in {0,1}
- AI-informed traders have superior predictive ability
- Uninformed traders have heterogeneous participation values z
- Competitive market makers price orders conditional on direction

Pricing and participation form a fixed-point problem

$$s = \frac{\alpha}{2[\alpha + (1-\alpha)(1-F(s))]}$$

- s: equilibrium half-spread
- alpha: AI penetration
- F(s): share of uninformed traders priced out

A wider spread reduces participation and makes order flow more informative.

More AI makes every order more revealing to the market maker

The gap between buy probabilities across states equals the AI penetration rate alpha.

- AI traders buy when V = 1 and sell when V = 0
- Uninformed traders trade only when z exceeds the spread

Takeaway

Higher alpha raises adverse-selection pricing pressure directly.

Higher AI penetration shifts the spread schedule toward collapse

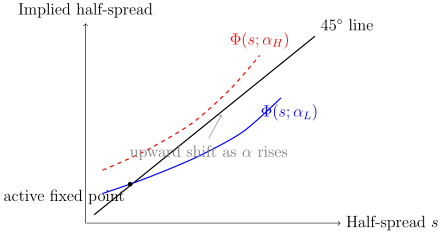

Figure 2: Schematic fixed-point representation. As AI penetration rises from α_L to α_H, the implied-spread operator shifts upward. The figure is illustrative only and does not impose a particular global functional shape beyond the comparative-static logic emphasized in the text.

Figure: Schematic fixed-point representation from the paper.

AI penetration above the threshold eliminates active equilibrium

Result

If alpha > 2 z-bar, no active equilibrium exists and the collapsed equilibrium is unique.

Intuition: The minimum AI-induced spread exceeds the strongest non-informational participation value.

The gradual phase is wider spreads and fewer ordinary participants

Before collapse, equilibrium spreads rise and participation falls with AI penetration.

- The spread operator is increasing in s and alpha
- Participation is t(alpha) = 1 - F(s(alpha))

Takeaway

The collapse theorem is the endpoint of a continuous deterioration mechanism.

AI rents are transfers, while participation creates baseline surplus

In the baseline welfare criterion, AI does not create direct productive surplus.

- AI rents equal spread losses borne by uninformed traders
- Market makers earn zero expected profit
- Real surplus comes from non-informational participation values

Takeaway

Private profitability and social value need not move together.

Participation-based welfare declines with AI penetration

$$W(\alpha) = (1 - \alpha) \int_{s^*(\alpha)}^{\bar{z}} z \, dF(z)$$

- W(alpha): baseline social welfare
- s*(alpha): selected active-equilibrium spread
- The integral captures surviving participation value

AI reduces welfare by shrinking the set of traders who remain active.

Private AI adoption can exceed the planner's optimum

Result

With non-negative adoption costs and the baseline welfare criterion, the planner chooses the lowest feasible adoption level.

Intuition: Adopters capture private rents but do not internalize the participation externality imposed on other traders.

Market design can target participation rather than price accuracy

- Participation subsidies reduce the effective cost of trading
- AI adoption taxes reduce privately chosen adoption
- Public information can reduce the AI informational edge
- AMM liquidity support addresses the analogous withdrawal spiral

Extensions preserve the participation feedback

- Imperfect AI replaces alpha with effective informed intensity
- Heterogeneous quality preserves the adverse-selection feedback
- Dynamic persistence creates path dependence and hysteresis
- AMMs and multi-market settings retain the core participation force

The theory yields testable predictions but not current-market evidence

The paper identifies empirical implications rather than claiming that prediction markets have already collapsed.

- AI exposure should predict wider spreads
- Wider spreads should reduce less-informed participation
- Order flow should become more concentrated in informed traders

Takeaway

Whether real markets are near the threshold remains an open empirical question.

AI creates an efficiency-survival trade-off

Takeaway

AI may improve price discovery while destroying the participation base that makes price discovery possible.